Chapter 7
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Displaying the Map Server Data using Arc-GIS server JavaScript API

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ABSTRACT

Applications for the web, desktop, and mobile are developed using the Arc-GIS server. Applications and services for managing, visualising, and analysing geographic data are offered by Arc-GIS for servers. The work makes use of the Popup-layout API reference and also makes use of two of its features as necessary. Each programming interface has features that help clients connect to geospatial information more easily. Keen web maps are a remarkable tool for visualising complex data. They help to break apart seemingly unrelated data, find hidden examples, and mine massive databases. Information can be created on the work surface, uploaded to the cloud, and shared online using the Arc- GIS server. The objectives of the work deals with exhibiting of geo-spatial attribute data using the facility of Java script application programming interfaces (API's) from Arc-GIS server. For a given Layer or Graphic, a PopupTemplate prepares and defines the content of a Popup. When a feature in the view is selected, the user can use the PopupTemplate to access values from feature attributes and values returned from Arcade expressions.

Keywords: Arc-GIS server; java script; attribute data; popup template; application programme interface.

1. INTRODUCTION

Arc-GIS (geographic information system) server is a complete and integrated GIS server. It gives the client the ability to share manuals, models, and tools with people both inside and outside of the association in a way that complements their work processes.

All aspects of crisis management, including reaction, preparation, recovery, and mitigation, depend heavily on geographic information and technologies. Since chipped in data is state-based and lacks the assurances that authoritatively produced information does, information quality is a major concern. During crisis

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time is the pith, and the dangers related with chipped in data are regularly exceeded by the advantages of its utilization is thoughtfully affecting overseeing information, organizingwork processes [1].

The main problems of Geographic Information Systems (GIS) are the large amount of information and performance of relational databases (RDBMS) in storage and in unstructured spatial data queries. This problem is significant, since the efficiency of accessing large amounts of spatial data over the Web today is important for conducting on-demand and real-time queries [2-4].

Web-GIS is utilized to gather, make, and convey geographic data wilfully by people in locales, for example, Wikimedia and open-road map for enabling people to make worldwide fix work of geographic data [5].

The geospatial-web, across association, and communicating with similarly invested people in successful networks. A portion of the empowering advancements for the geospatial-web are geo-programs, for example, google earth, NASA world wind. These three-dimensional stages change the utilization and creation of media items. They not just unite individuals of comparative interests, perusing conduct, yet additionally uncover the geographic dissemination of webadministrations, assets [6].

In all parts of crisis oversight, geospatial information and apparatuses can possibly help spare lives, limit harm, and decrease the expenses of managing crises.

The achievement of any innovation is as much about the human frameworks in which it is installed as about the innovation itself [7].

It is conceivable to progressively gather application from various GIS web administrations for use in an assortment of customer applications. ESRI offers a worker-based stage from which geographic data is distributed and spread inside associations, across authoritative limits, and to people in general. Arc-GIS server is ESRI's essential worker GIS item. It is a finished and incorporated worker-based GIS that accompanies out of-the-crate, end client applications and administrations for spatial information perception, spatial examination and the executives [8].

Arc-GIS online is the cloud version of Arc-GIS enterprise. Arc-GIS Online is a cloud-based mapping and analysis solution. It is used to make maps, analyse data, and to share and collaborate Bhatia et al, ARC- GIS server, the web application item from ESRI, can be upgraded by building custom web administrations. To get past Arc-GIS worker benefits, a scope of customer applications and engineer structures and APIs like Arc- GIS desktop, rich Internet applications, for example, the Arc-GIS API for Microsoft Silverlight/WPF, and web applications for example, the web application development framework (ADF) for Java/Microsoft.NET and server object extensions are given by ESRI [9].

Arc-GIS worker additionally give a wide scope of API's based on guidelines which permits customers to play out the customization of geo-spatial information relying upontheir requirement [10].

It is a wide and fused programming for building operational GIS. Arc-GIS contains four programming parts, segments for capacity and overseeing geographic information in documents and databases; a geographic data structure for displaying part of this present reality; a lot of out-of-the-crate applications for making, controlling, altering, dissecting, planning and to spread geographic data; and an assortment of web benefits that give substance and abilities to arranged programming customers. Portions of the Arc-GIS programming framework can be sent on mobiles, and personal computers. From the end client perspective Arc-GIS has wide running functionalities bundled up into a conventional arrangement of menu driven GIS applications that actualize key geographic work processes [11].

The PopupTemplate class has title and content properties that serve as a template for converting a feature's characteristics to HTML.The parameter substitution syntax is fieldName or expression/expressionName.

When you click on a Graphic, the default behaviour is for the view's Popup to appear. This default behaviour necessitates the use of a PopupTemplate.

With the fieldInfos property of PopupTemplate, you can format Number and Date field values and override field aliases. Users may also be able to execute actions connected to the feature, such as zooming in on it or running a query based on the feature's location or properties, by adding actions to the template.

Custom actions may be included in PopupTemplates. When these actions are activated, the developer's custom code is executed. For further information, see the actions property [12].

2. BENEFITS OF ARC-GIS SERVER

Publish quick, keen web maps customized to the crowd, significantly reinforcing business and asset choices with continuous geo-knowledge.

Centrally deals with the geo-information, giving better information security and trustworthiness for a large portion of the significant data resources.

Simplify access to enormous volumes of symbolism assets, broadly diminishing capacity expenses and information handling overhead.

3. WORK FLOW OF THE PAPER

3.1 Accessing the Webpage for Displaying the Base Map

Using the HTML base map is displayed with the property: base map. The base map property is having different qualities like: topo, streets, satellite, hybrid, dark-Gray, Gray, national-geographic and so on. In light of the reliance any worth can be picked base-map property. At that point in the wake of utilizing the map see property the base map can be shown. Map-view is additionally having different properties like animation, background, map, container, center, zoom, navigation and so on which can be gotten to on the reliance.

Base map is accessible under "Esri/Map" for displaying the map, view is required it is accessible under “esri/views/MapView". The output of the map view is appeared in result segment with (Fig. 1). The latitude and longitude esteems are given to center property. Here for this situation Hyderabad's latitude and longitude esteems are given. So, Hyderabad area will get engaged in the output.

3.2 Displaying the Services in Map View Environment

Various services are gladly available like map service, feature service, image service, geo-processing and geo- coding services.

Map services are the one analogous to MXD document of Arc-Map. It consists of many feature services in sole mapservice.

Feature service will focus on one of the particular features like point, line, polygon.

3.3 Map Image Layer and Feature Layer

They are available under the API’s- "esri/layers/MapImage Layer" and "esri/layers/Feature Layer " API reference.

Map image layer allows to analyse data from sub layers defined in a map service, export images instead of features.

Map service images are robustly generated on the server based on a claim, which includes a bounding box, level of detail, spatial reference and other options.

The output of map image layer is shown in result section with (Fig. 2). It contains the information about total layers of sample census data.

A feature layer is a single layer that can be created from a map service or feature service.

URL: It is a common property of both map image layer and feature layer. The value for URL property can be identified from Esri/layers sections. All the URL’s are mentioned at:
“https://sampleserver6.arcgisonline.com/arcgis/rest/” in the services section. Same URL can be specified to feature layer or map-image layer.

The output of feature layer is shown in result section with (Fig. 3). It contains the information concerning about layer only [6] If the same task is done with map image layerit gives the output of all the layers.

4. POPUP TEMPLATE

It is realistic under "esri/PopupTemplate" API. A popup template characterizes and arranges the substance of a popup for a graphic or explicit layer. A popup template permits the client to get to information from include characteristics and information came back from arcade terms when an element in the view is chosen.

It gives insights regarding showed highlight by tapping on it. Its conduct changes with zoom level of yield map and tapping on it. A popup template permits the client to get to an incentive from include qualities of specific element in output. It contains various properties like actions/content/title etc relying upon the necessity anyvalue cane be opted that match the criteria.

Popup template is having various properties like actions, content, title and so on. Based on dependency 2 properties content and title have been used in current work.

Content: It can be accessed in five different ways namely content, string, function, promise, auto cast. In the existing work it has been used in string format.

The content property can be accessed in different ways
i.e. attribute fields can be assigned one or multiple at atime.

Ex content: "CLASS:{CLASS}
 POP2000:

{pop2000}
 ObjectID:{OBJECTID}"

In the output (Fig. 4) user will be able to see the information reg only the attribute data mentioned in content section. If the user wants complete attribute data to be displayed "*" is used.

Ex content: "{*}".

In Fig. 5 total information of attribute data is displayed on the output screen, the only difference between Figs 4 and 5 is in Fig. 4 only few properties like class, pop 2000 and object ID is getting displayed where as in Fig. 5 total attribute data is getting displayed because of "*".

String: A pop-up's content can be a easy text or string value indicating field values or Arcade expressions. Expressions must be defined in the expression-info's property.

Title: Title can be accessed in three different ways namely string, function, promise. In the current paperTitle has been used in string format.

Ex title: "Information in {AREANAME}". In the output (Fig. 4) user can able to see the title in the prescribed format i.e. information in Bangor. Whereas AREANAME is the attribute data associated with user selection on Image layer.

5. DISCUSSION

A sample base map using the Arc-GIS server API configuration is shown below (Fig. 1). The view set was "streets" in this code. For altering the subject, several attributes such as satellite, hybrid, and so on can be used.

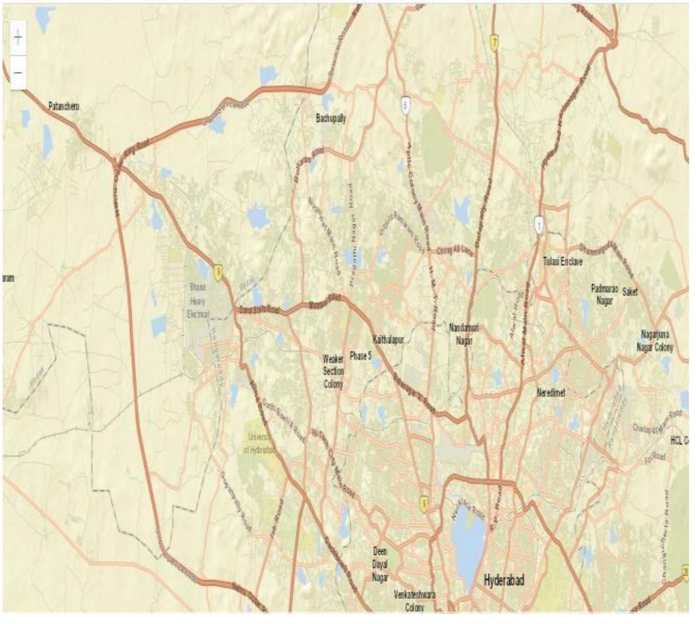

Fig. 1. Base map of latitude longitude provided byuser

The figure below (Fig. 2) shows complete layers of vector data. The map image layer is used to access all of the layers at once. The parent of the feature layer is believed to be the Map Image layer. There are three unique layers in the above output: point information, line information, and polygon information. below (Fig. 2) shows complete layers of vector data. The map image layer is used to access all of the layers at once. The parent of the feature layer is believed to be the Map Image layer. There are three unique layers in the above output: point information, line information, and polygon information.

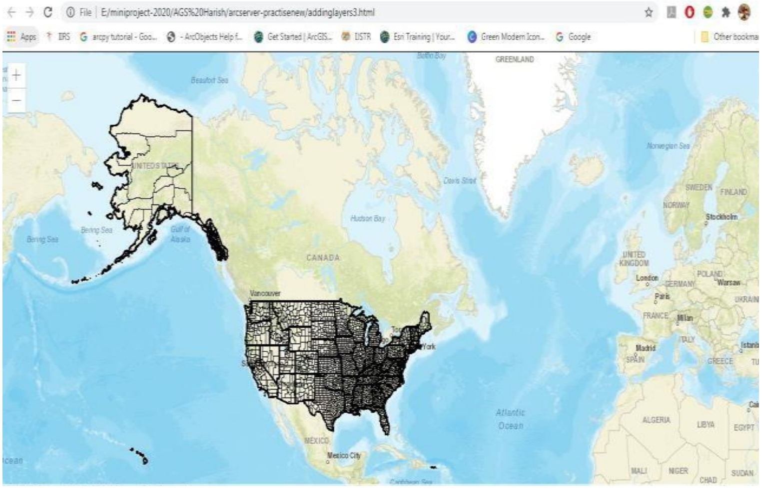

Fig. 2. Map image layer containing all the 3 layersof sample census data

The polygon layer of census data is shown below (Fig. 3). Using the Feature layer, it is possible to depict different layers such as line and point information. The feature layer communicates with just one layer at a time.

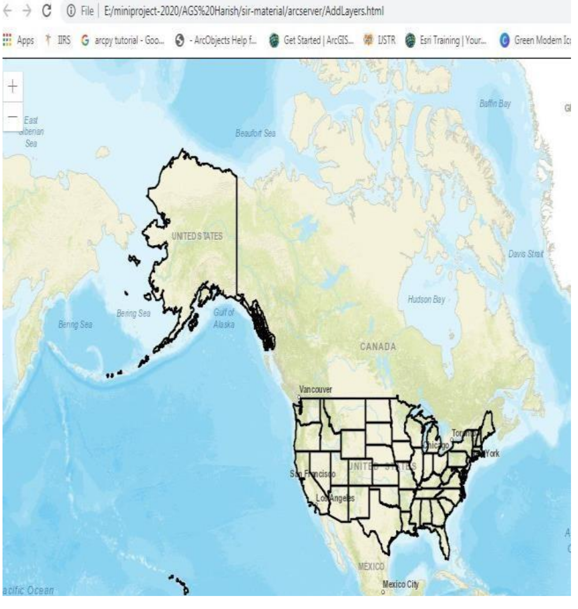

Fig. 3. Feature layer containing layer 3 of samplecensus data

When one of the elements is selected, its details are displayed in a small window, as seen in (Fig. 4). This is accomplished using a popup template reference. The issue with the Query task is that it takes more time to learn the intricacies of the highlight feature. This can be reduced with the popup layout.

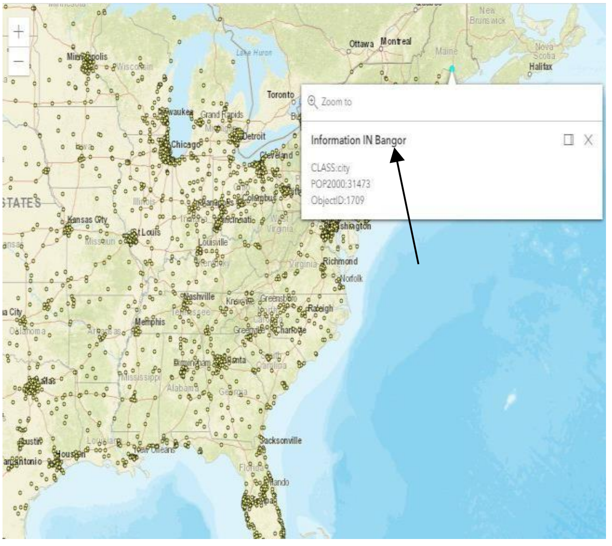

Fig. 4. Popup template for displaying the featureinformation

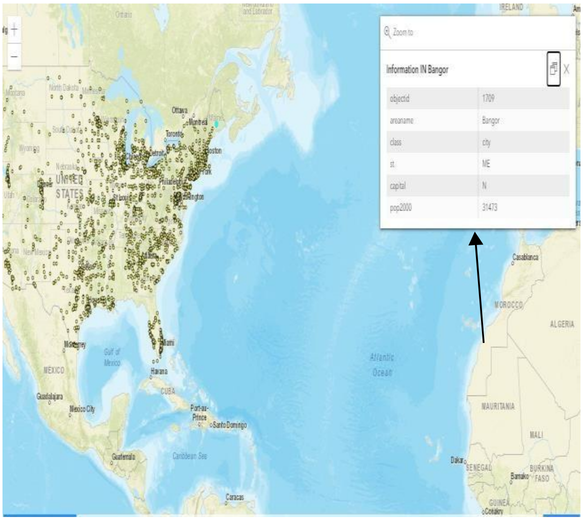

Fig. 5. Popup template for displaying the completefeature information

Because of the "*" value applied to the content property, the following (Fig. 5) shows complete attribute details of user selection.

6. CONCLUSION

The popup-template is used to display details of selected features based on the zoom level selected by the user. Popup-layout is a technique for retrieving the details of any vector data with a single click on the chosen highlight. The properties of the popup template, such as content or title, have been employed based on the user's reliance.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

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Biography of author(s)

Dr. Ballu Harish
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He is an assistant professor since January 2016 in JNTUH-Institute of Science and Technology, Hyderabad, India. He has Six years of experience as Assistant professor in Center for Spatial Information Technology, IST-JNTUH campus, India and an Associate Photogrammetry in Avienon India Pvt Ltd from March-2015 to January 2016. He has Completed a Mini R &D project ttitled " Exchange of Geospatial data through Geoweb" sanctioned by JNTUH IST TEQIP-III for a period of 1 Year. He was a Co-cordinator of Environment models in Geospatial domain(2020), International GIS day(2020), Role of Remotesensing in environemnt studies(2020), Geomatics for watershed management(2020), Expert lecture on Mobile GPS apps(2020), Expert lecture on Arc server(2020), Three days workshop on Arc-objects(2021), Three days workshop on QGIS(2021), Two days workshop on ILWIS(2021). He is Certified in Microsoft Digital Literacy Test on March-03-2011, IIRS 18th Outreach program "Basics of Remote Sensing, Geographical Information System & Global Navigation Satellite System" conducted by CIST-IST-JNTUH during August-November 2016, IIRS 19th Outreach program "Remote Sensing and GIS Applications in Carbon Forestry" conducted by conducted by CIST-IST-JNTUH during February 16, 2017 to March 10, 2017, IIRS 20th Outreach program '"Microwave Radar Remote Sensing and its Applications" conducted by CIST-IST-JNTUH from April 10, 2017 to April 27, 2017. His Research Area is Geospatial Technology. He has Young Researcher Award-InSC Banglore in his credit. He also Working as Reviewer for International Journal (Journal for Educators, Teachers and Trainers- JETT. He has 32 published papers in national and international journals.

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